

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): **Model:** $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N : population at time t ; K : carrying capacity (depends on resources); $r, K > 0$.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \xrightarrow{t \rightarrow \infty} K$ **Equilibrium points:** $N = 0$ (unstable), $N = K$ (stable). **Quickest:** $N = \frac{K}{2}$ ($<$: 凹; $>$: 凸)

Delay Model: **Model:** $\frac{dN}{dt} = f(N(t), N(t-T))$ T : delay time > 0 周期解 | 与初始值无关 ps : To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ ²**Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)

^{1,2} **Periodic sol:** stable limit cycle $N(t + t_p) = N(t)$, t_p : period. **Amplitude:** rT **Equilibrium Points:** $N = 0 | N^* = 0$ (unstable); $N = K | N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): **Model:** $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ $r, K, E > 0$; EN : harvesting rate/unit time; E : effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) **Yield:** $Y(E) = EN$ $\cdot E > r$: die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) **Steady State:** $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y_0): **Model:** $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0 : constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** $0 < Y_0 < Y_M$: Two positive points (unstable[小] and stable[大]). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: **Verhulst:** $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r \leq 0$, x^* unique. If $r > 0$, $|h| > h_c$, x^* unique; $|h| < h_c$, x^* three. (两边 stable, 中 unstable). Bifurcation points: $h = \pm h_c = \pm \frac{2r}{\sqrt{3}}$. **cusp point:** $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

Stability	Stable: $f'(N^*) < 0$		Unstable: $f'(N^*) > 0$		Stable: $ \lambda < 1$	Unstable: $ \lambda > 1$ (chaotic)	Periodic: $ \lambda = 1$ $\begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$
Linearise	$N(t) = N^* + n(t), n(t) \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$				Let $N_t = N^* + n_t, n_t \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$		
Bifurcation \\\nPerturbations\nat N^* Periodic	Saddle-Note (Normal Form) $\dot{x} = r \pm x^2$ $0 \rightarrow S \rightarrow S$	Transcritical $\dot{x} = rx - x^2$ $S \rightarrow U \rightarrow S$	(super-)Pitchfork $\dot{x} = rx - x^3$ $S \rightarrow S \rightarrow U \rightarrow S$	Sub-critical Pitchfork $\dot{x} = r + x^3$ $U \rightarrow U \rightarrow S$	$\lambda < -1$: $\uparrow$ oscillations If $fP(N^*) = N^*$, N^* is p -periodic.	$-1 < \lambda < 0$: $\downarrow$ oscillations $\forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic	$0 < \lambda < 1$: $\downarrow$ Monotone Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists.
Other	Time-Independent: At N^* : $\frac{dN}{dt} = 0$ $N(t) = N(t-T) = N^*$ $f(N^*) = 0$				Max/Min: (如果单峰 (unimodal)) 2-periodic $f'(N_m) = 0$, $N_{\max} = f(N_m)$; $N_{\min} = f(N_{\max})$		
Other	Tangency Condition: If $\dot{x} = f(x, r)$, 分岔点 (x^*, r_c) : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$ 两点合并: coalesce .						

Bifurcation: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt} \Rightarrow$ Assume $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$ Find $\lambda = g(\lambda) \Rightarrow$ Let $\lambda = u + iw \rightarrow u = g_1(u)$ and $w = g_2(w) \Rightarrow u < 0$ asy.stable; $u > 0$ unstable. **Find T_c :** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T = 0$ stable. $\Rightarrow$ Using $u = 0$ (i.e. $\lambda = iw$) to find T_c .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a - bP)$; $\frac{dP}{dt} = P(cN - d) \Rightarrow$ nondim: $\frac{du}{d\tau} = u(1 - v)$; $\frac{dv}{d\tau} = \alpha v(u - 1)$ **Equilibrium:** $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H$, $H > H_{min} = \alpha + 1$ constant. (closed phase) **Conservation Law:** $\frac{dH}{d\tau} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u, v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$ If $+/-$ definite (i.e. $\forall \lambda_i > 0 (< 0)$)

$\Rightarrow$ It's local min/max at (u^*, v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$ at equilibrium (u^*, v^*) . Let $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$, then $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$ Find eigenvalues λ of J by $\det(J - \lambda I) = 0$

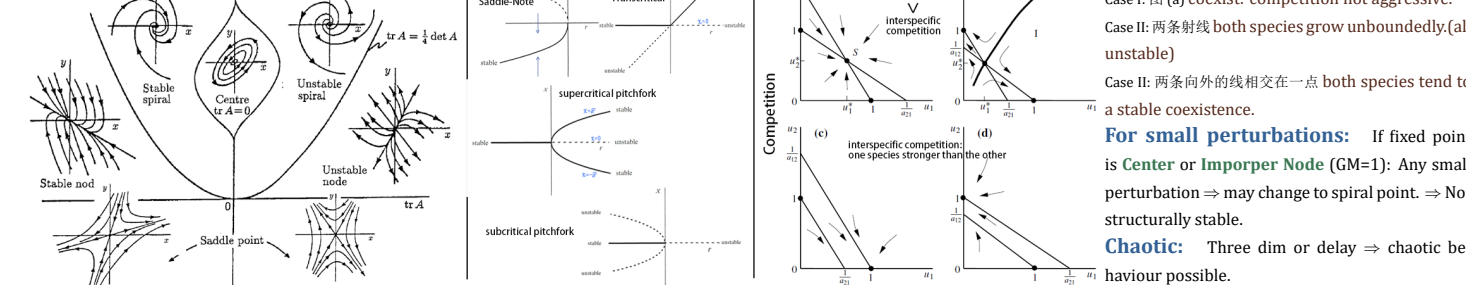
¹ $\det(J) > 0$, $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$ stable node $\text{Im}(\lambda) = 0$; stable focus=spiral point $\text{Im}(\lambda) \neq 0$; ² $\det(J) > 0$, $\text{tr}(J) > 0 \Leftrightarrow \Re(\lambda) > 0 \rightarrow$ unstable node $\text{Im}(\lambda) = 0$; unstable focus=spiral point $\text{Im}(\lambda) \neq 0$

³ $\det(J) < 0 \Leftrightarrow \text{Im}(\lambda) = 0$, $\lambda_1 < 0 < \lambda_2 \rightarrow$ saddle point (unstable); ⁴ $\det(J) > 0$, $\text{tr}(J) = 0 \Leftrightarrow \lambda = \pm iw$ (pure imaginary) $\rightarrow$ center (neutrally stable)

Null clines: $f(u, v) = 0$ or $g(u, v) = 0 \rightarrow$ 得 u, v 表达线条是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有边界的外法线方向 $\mathbf{n}$ 在对应边界上的点, 满足 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be < 0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (具体四种情况看图)



3 Reaction Kinetics

Enzyme Kinetics: e.g. $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$ K_i : rate of reaction; S : substrate; E : enzyme; SE : complex; P : product.

Singular Perturbations Analysis: For $\frac{du}{d\tau} = f(u, v)$, $\varepsilon \frac{dv}{d\tau} = g(u, v)$, $0 < \varepsilon \ll 1 \Rightarrow v$ fast (when $g \neq 0$), u slow. **Quasi-steady-state approximation:** $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

Taylor: $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$; $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$ **Outer solution:** Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1)|\varepsilon = 0$ Valid for $\tau > 0$ (slow time); $v_0(0)$ invalid. $\rightarrow$ not uniformly valid for $\tau \geq 0$

Inner solution: $\sigma = \frac{\tau}{\varepsilon}$, $U(\sigma; \varepsilon) = u(\tau; \varepsilon) \Rightarrow \frac{dU}{d\sigma} = \varepsilon f(u, v)$, $\frac{dV}{d\sigma} = g(u, v) \rightarrow$ Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1)|\varepsilon = 0$ (known as **boundary layer** ($\tau = 0$ 处)); Valid $\tau = 0$ (fast time)

Asymptotic Matching: $\lim_{\sigma \rightarrow \infty} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$ **Uniformly Valid Solution:** $u(\tau; \varepsilon) = u_0(\tau) + \mathcal{O}(\varepsilon)$; $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + \mathcal{O}(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + \mathcal{O}(\varepsilon), & 0 < \tau \ll 1 \end{cases}$

Activator-Inhibitor Model:

4 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor:** $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$ **Taylor Expansion:** $f(x, r)$ at (x_0, r_0) : $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ **Linerisation:** $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$ $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$

Hyperbolic Functions: $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$